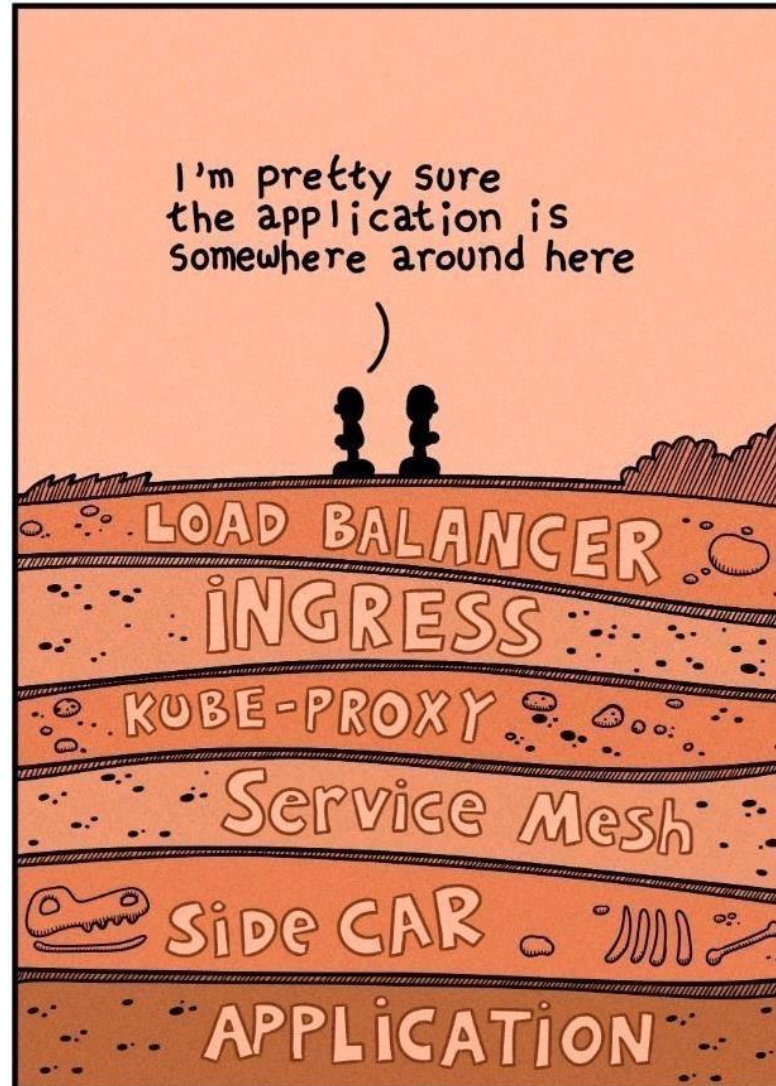


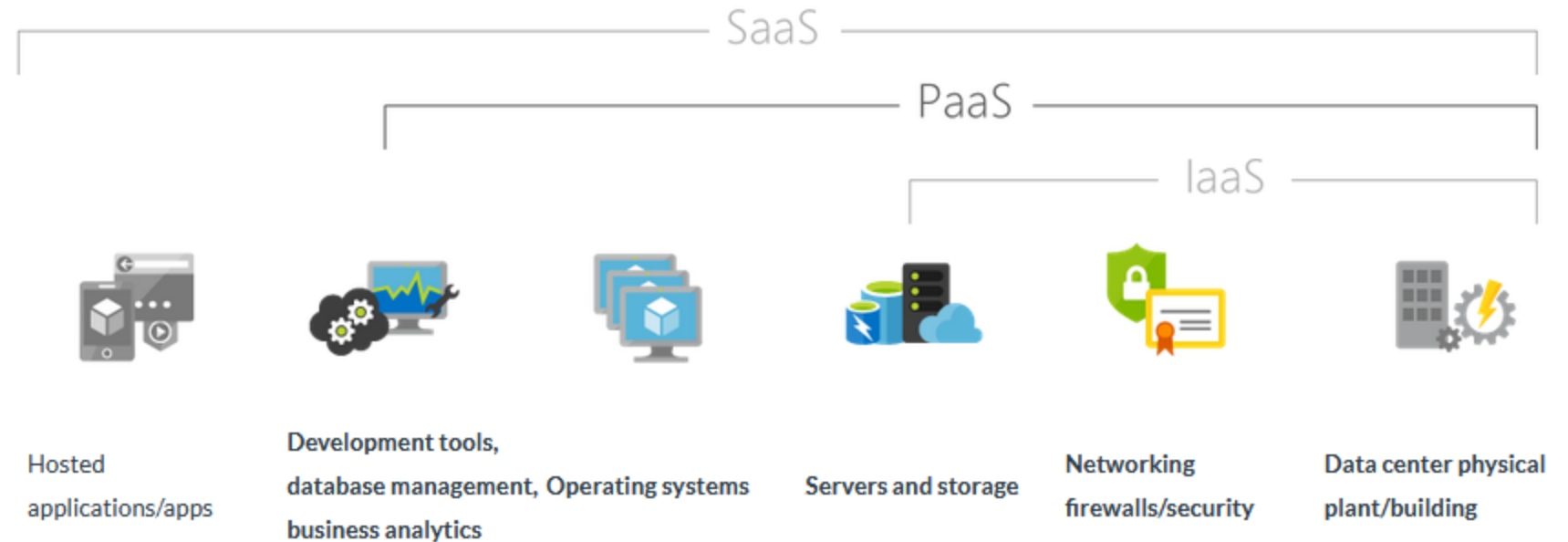
Public Cloud Services PaaS auf Hyperscaler



For the start...



PaaS – Platform as a Service



Use Cases:

- Operating of own business code, (often containerized applications)
 - Analytics und Business Intelligence
 - Authentication, Workflow Engines, Directory Services, etc.

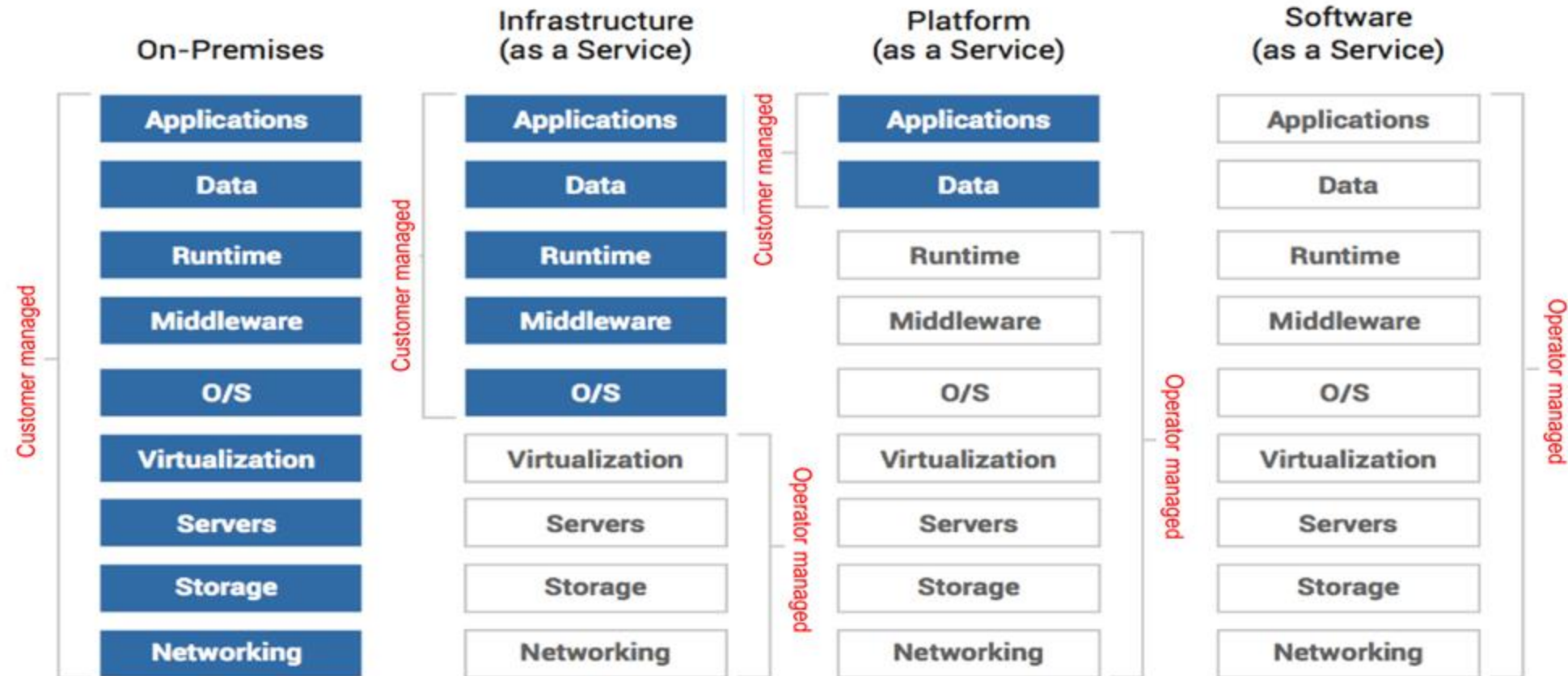
What functionalities do you expect from a PaaS?

NIST Definition of PaaS

The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using **programming languages, libraries, services, and tools supported by the provider**. The consumer does **not manage** or control **the underlying cloud infrastructure** including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

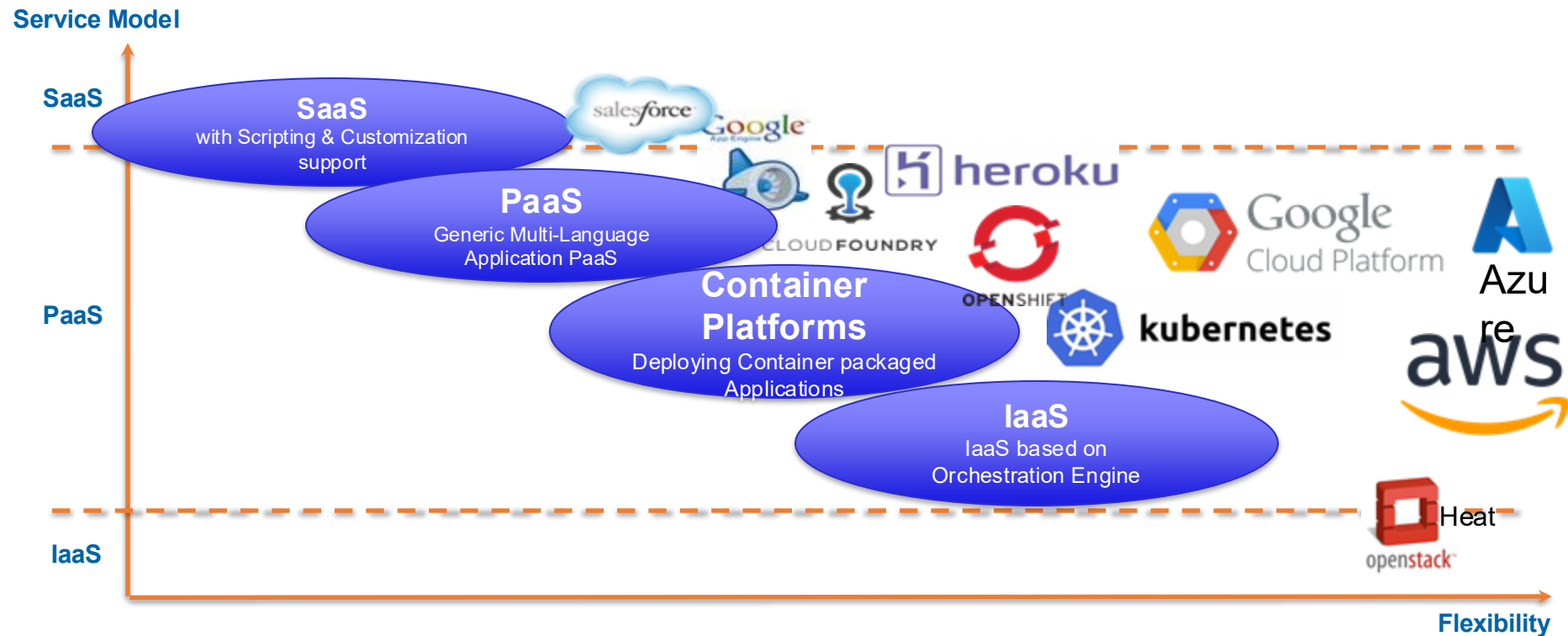


What are components of a PaaS?



What is the runtime?

Blurred lines....



Container Platform VS PaaS

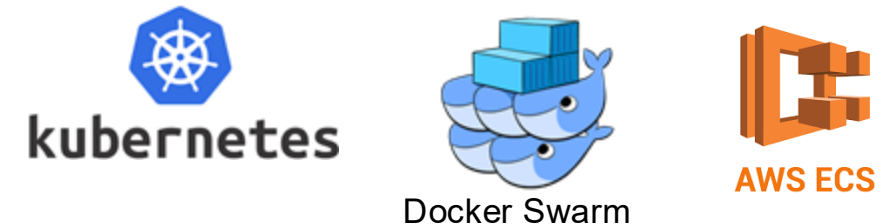
Application (PaaS / Application Platform)

- You are in charge of the application code only
- “Here is my code, run it. I don’t care how”



Container Orchestration

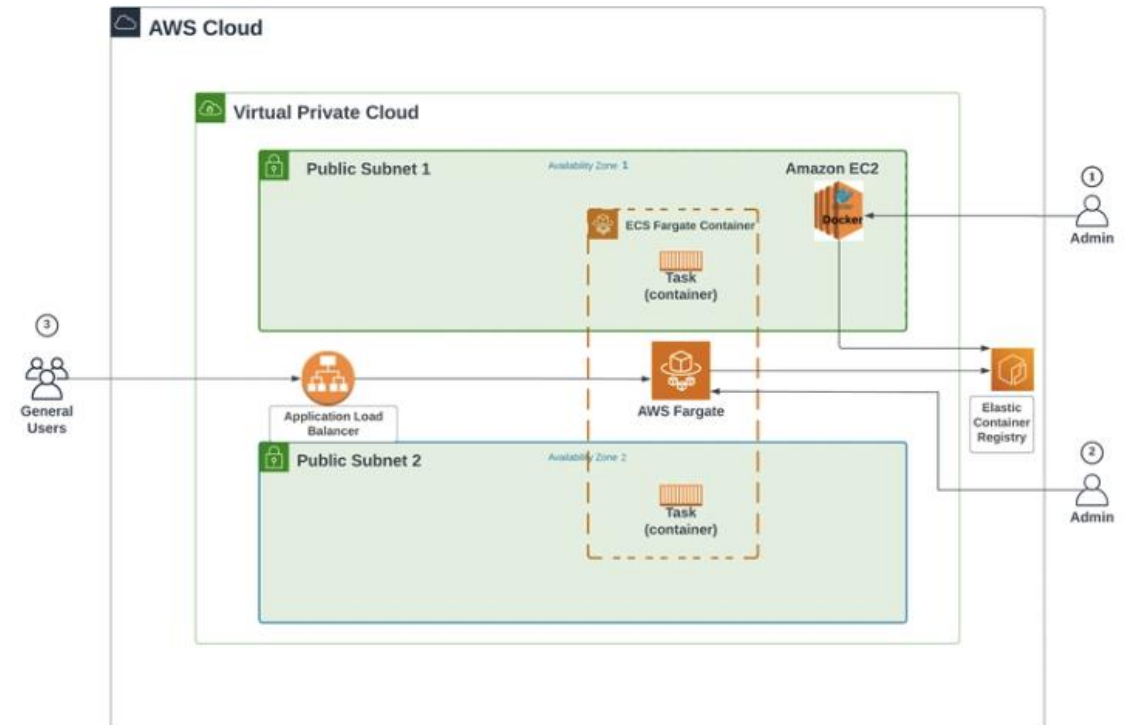
- You are in charge of a container including its runtimes
- “Here is my container, I tell you how you run it”



Serverless Compute

Given the case, your workload is an OCI-Container...

- You either have the ability to schedule it via EC2 (IaaS with all Pros / Cons)
- ...or to schedule it serverless (Fargate / Azure Container Instances)



The price of serverless...

Number of tasks	vCPU per task	m5.xlarge number of vCPU available	CPU reservation rate	Memory per task (GB)	m5.xlarge memory available	Memory reservation rate
8	4	32	$4 \cdot 8 / 32 = 100\%$	16	128	$8 \cdot 16 / 128 = 100\%$
8	4	32	100%	8	128	50%
8	2	32	50%	8	128	50%

Fargate generates less operational burden, pricing is fixed to vCPU and Memory per Task per Second.

<https://aws.amazon.com/blogs/containers/theoretical-cost-optimization-by-amazon-ecs-launch-type-fargate-vs-ec2/>

The price of serverless



Besides Compute...

Storage:

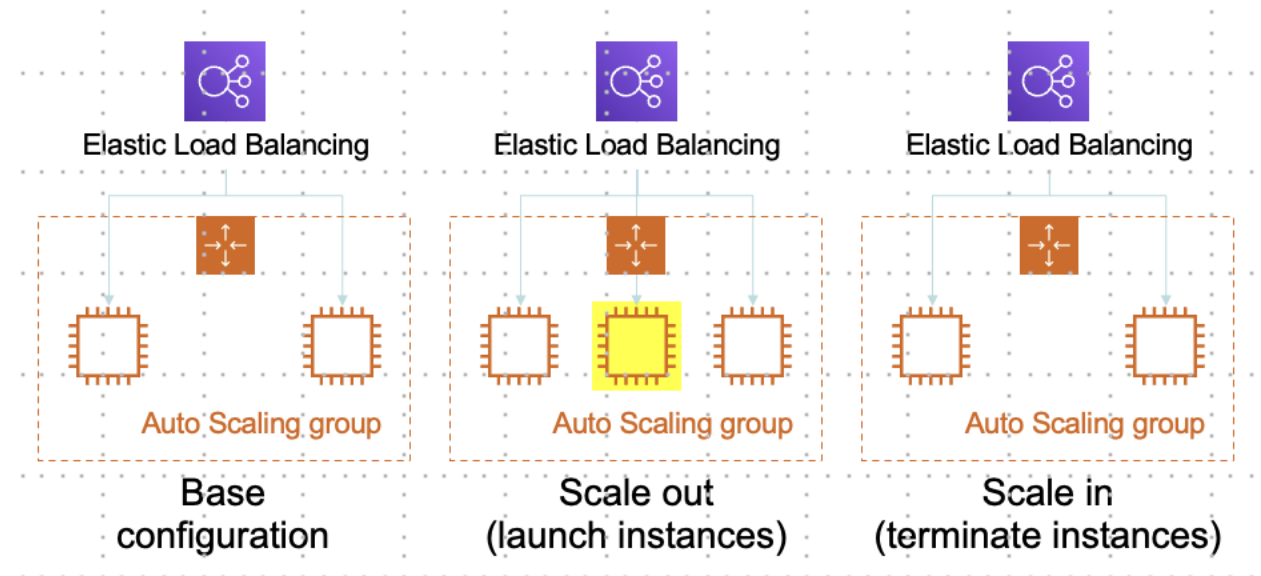
- Database
- File

→ To be covered in the next weeks

Load Balancing, Ingress:

- OSI-Level 3/7
- Must recognize the Infrastructure (scale in/scale out)

→ Also to be covered in the next weeks



AWS Beanstalk

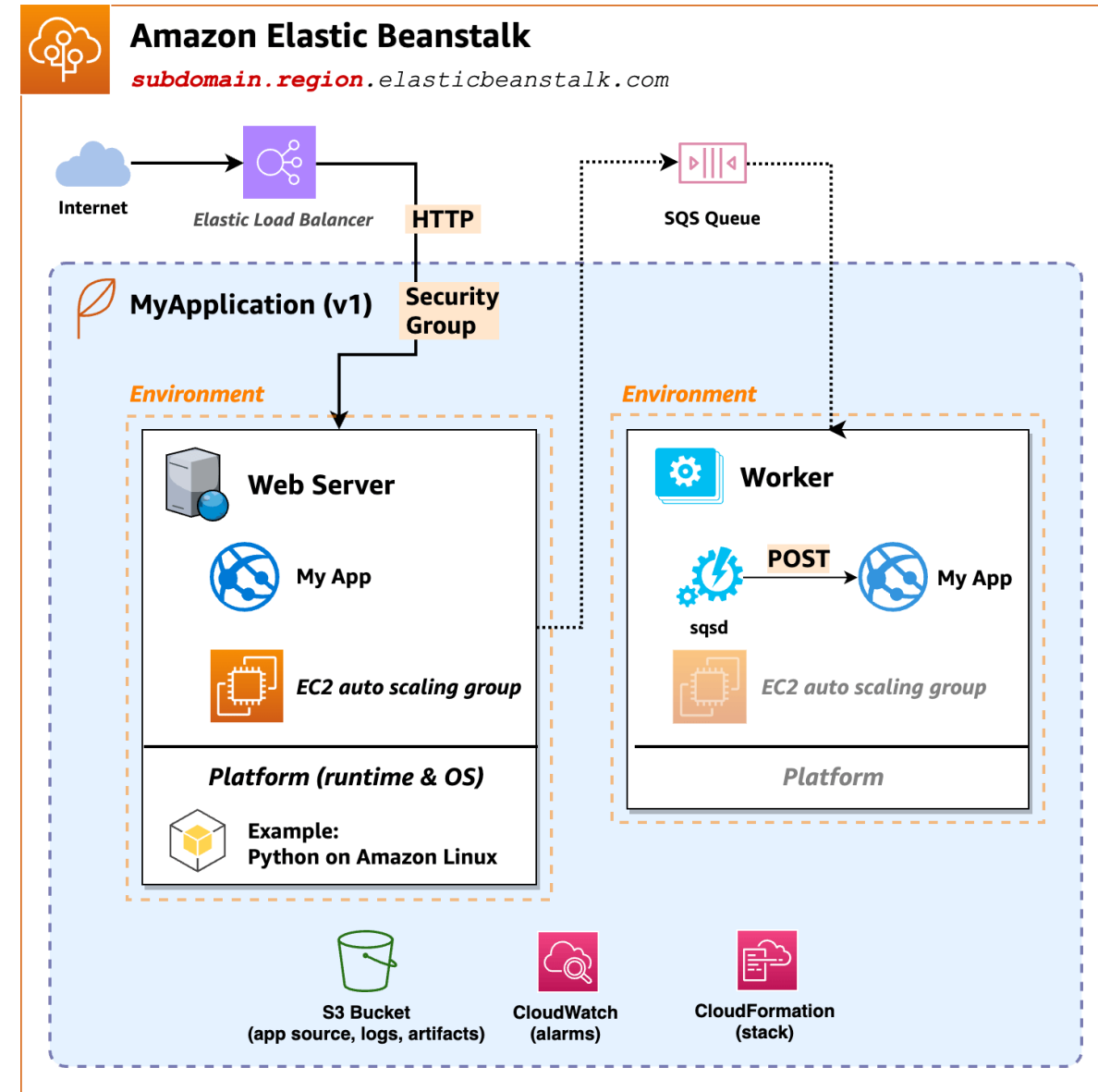


PaaS-Offering from AWS:

- More or less an orchestrator for different services including
 - Load Balancing
 - IaaS (or Container Orchestration) including Autoscaling
 - Database
 - Monitoring

→ Services are operated by AWS

<https://docs.aws.amazon.com/elasticbeanstalk/latest/dg/Welcome.html>



Feedback regarding Complexity in Setup / Operating

AWS ECS

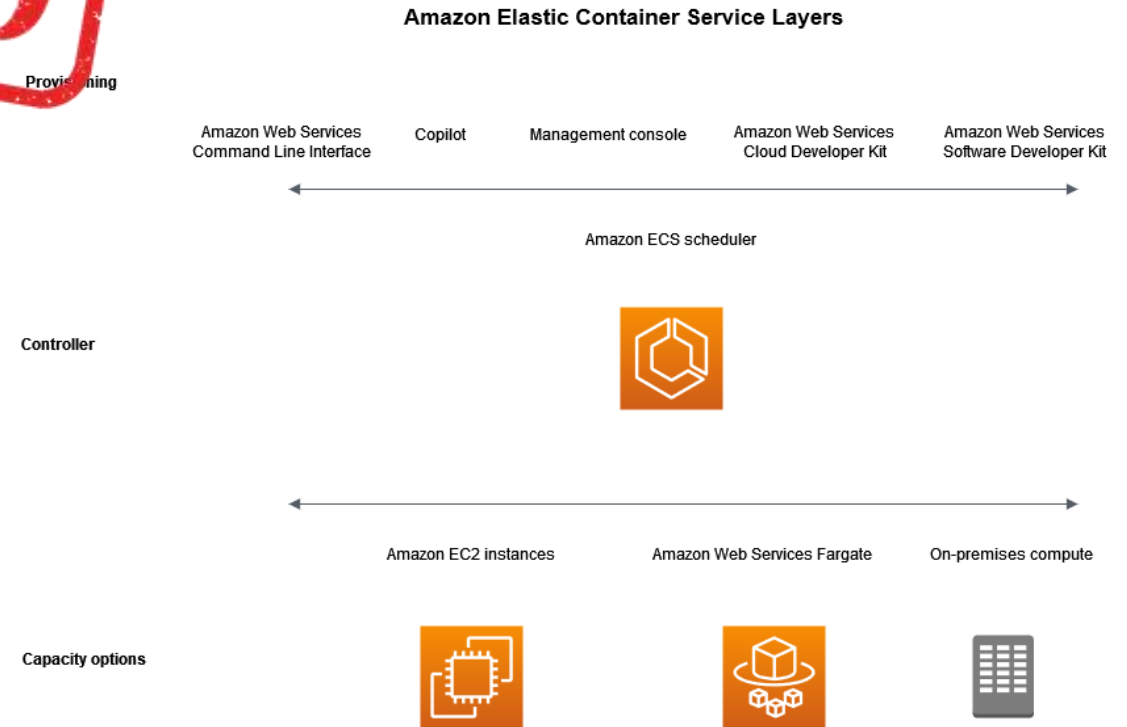


Container-Hosting from AWS:

- Cares about the lifecycle of a set of container (called task)
 - Scheduling as service needs a “service”
 - LoadBalancing, Networking, Provisioning not included
- Can make use of EC2 or AWS-managed infrastructure

→ Services not necessarily operated by AWS

<https://docs.aws.amazon.com/ecs/>



Feedback regarding Complexity in Setup / Operating

Updates / Upgrades of PaaS/SaaS-Products

Users need to upgrade their underlying services regularly:

- Patch/Minor-Updates: might be automatic scheduled, might result in replacing existing pod/container/processes
- Major: must be enforced by user
- If update does occur, availability of service is not guaranteed

The screenshot shows the Microsoft Azure portal interface. On the left, the navigation pane is open, showing the 'Upgrades' section for the 'hs25-group-teacher-sysd' Kubernetes service. The main content area displays the 'Upgrade Kubernetes version' dialog. The dialog includes a search bar, a list of upgrade options, and a table of current cluster settings.

Upgrade	Automatic upgrade	Kubernetes Version	Upgrade scope	Force upgrade	Upgrade readiness
1.33.2	Disabled	1.33.2 (current)	Upgrade control plane + all node pools	☐	Check

Additional settings shown in the dialog:

- Fleet Manager: None
- AKS pricing tier: Free
- Automatic upgrade scheduler: No schedule set

The dialog also includes a 'Save' button and a 'Cancel' button. A note at the bottom states: 'It is highly recommended to run this check prior to your upgrade to identify any potential issues and take corrective actions'.

Lifecycle of PaaS/SaaS-Products

Services might be shut down with few months latency.

- Intransparent Roadmap upon canceling of service
- Customers are forced to find different platform
→ the dark side of vendor locking

Documentation > AWS IoT Events > Developer Guide

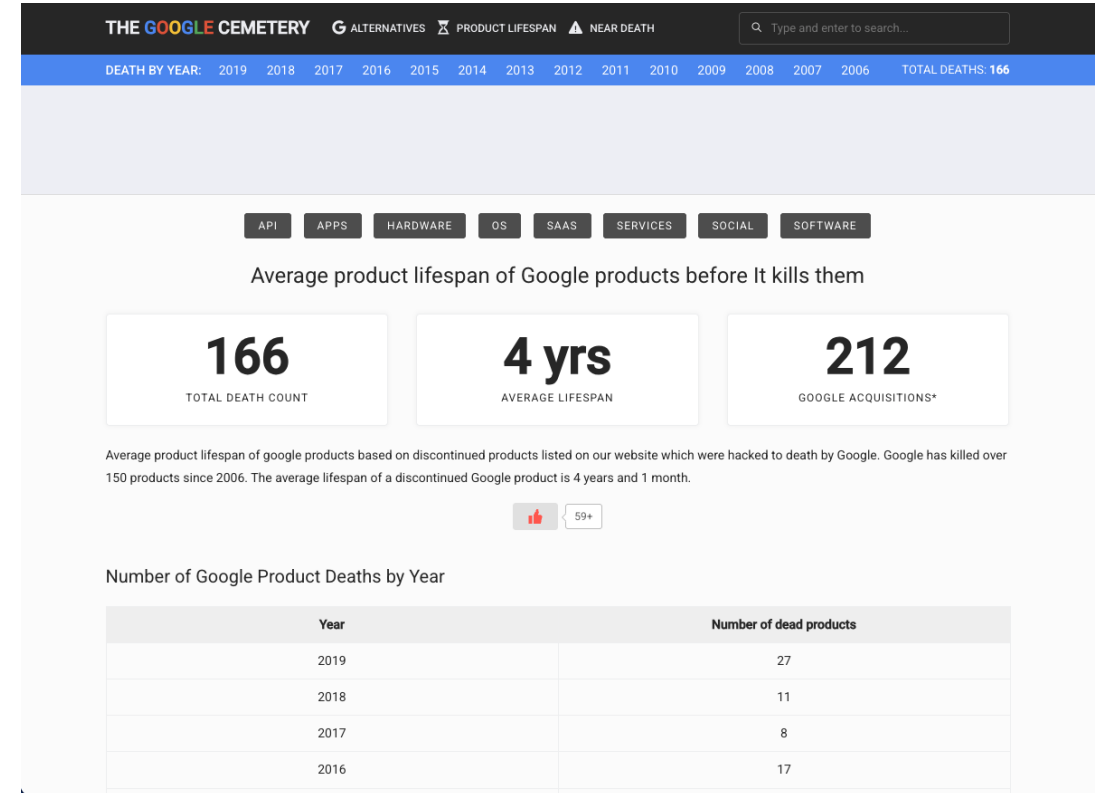
AWS IoT Events end of support

[PDF](#) [RSS](#) ☐ Focus mode

End of support notice: On May 20, 2026, AWS will end support for AWS IoT Events. After May 20, 2026, you will no longer be able to access the AWS IoT Events console or AWS IoT Events resources. For more information, see [AWS IoT Events end of support](#).

After careful consideration, we decided to end support for the AWS IoT Events service, effective May 20, 2026. AWS IoT Events will no longer accept new customers beginning May 20, 2025. As an existing customer with an account signed up for the service before May 20, 2025, you can continue to use AWS IoT Events features. After May 20, 2026, you will no longer be able to use AWS IoT Events.

This page provides instructions and considerations for AWS IoT Events customers to transition to an alternate solution to meet your business needs.



<https://gcemetery.co/google-product-lifespan/>

Optional Tutorials

- ECS Video and Beanstalk Tutorial:
<https://awsacademy.instructure.com/courses/137586/modules> → Modul 6

